

Kent Ramliden: The Fifty-five Year Journey of Venona Message No: 3/NBF/T1463

On 28 November 1941 the Soviet Military Intelligence (GRU) Station in Stockholm, Sweden sent yet one more message to Moscow Centre from its astonishingly well informed source known only as "K". This time it contained the weekly updates of casualty and related figures in the German units, and their Finnish allies, fighting the Red Army on the northern front. The message was very detailed, obviously based on an inside source with very good access, and we can all read it now because in 1996 the NSA released the message to the public as Ref No: 3/NBF/T1463 (T1463), part of Venona tranche number five. It turns out that the message was in all likelihood created a day or two before 11/28/1941, in a different place and language, and by the time of the NSA publication, it had been through at least six countries, been intercepted at least twice in a minimum of two languages, encrypted twice, once in an alphanumeric machine cipher and once in a code superenciphered with one-time pads. It had been broken cryptanalytically by two countries (one of which partially) and stolen by a spy working for a third. Despite the fact that we now only have the English translation of the final fragmented decryption available, it has been possible to reasonably trace the message via research into the various intelligence operations launched by Sweden, The Soviet Union and the UK/USA against the original German message. Here is the journey of an encrypted message from World War II.

On June 22, 1941 Germany attacked the Soviet Union, and almost immediately Finland signed on with the Germans hoping to take back territory lost to the USSR in the Winter War of 1939-40. Throughout the fall of 1941 strenuous, though unsuccessful, efforts were made to break through the Russian lines and take control of the strategic railroad running from Murmansk south toward the Russian heartlands, on which nearly all Allied relief supplies ran. In charge of the German northern front was Army HQ Norway (AOK Norwegen) which made heavy use of Finnish, Norwegian and Swedish telephone lines for their communications, and the message in question was part of the normal updates sent to Berlin HQ (OKW). These HQ-to-HQ messages were encrypted, generally with a high level German cipher machine known as the Geheimschreiber, (at Bletchley Park known as Tunny), more complicated than the Enigma. The fact that the traffic was travelling on land lines, out of reach of the big Allied intelligence services, made the Germans extra sure that their communications through Sweden were secure. They were wrong about that.

Small, but extremely capable, the Swedish Signals Intelligence organization FRA (predecessor) had broken the Geheimschreiber already in the summer of 1940 through the efforts of mathematician Arne Beurling. This man later got Einstein's old office at Princeton University. FRA had access to everything that ran on the telephone lines and much of what flew through the ether. So our message in question was generated and encrypted by the Germans in Northern Finland, likely on 27 November 1941. On its way south to the German High Command (OKW) in Berlin it was intercepted by the Swedes who immediately started decrypting it. By this time their Geheimschreiber decrypting process was so fast that they often knew the contents before the Germans. The FRA then used a decidedly low tech report distribution approach. They put the decrypts into a secure briefcase where only the FRA and the relevant recipient, in this case the Swedish Army HQ, had keys. They then sent the reports by a special courier off to deliver the briefcase contents. The Swedes considered this to be an efficient and completely secure method of report distribution. They were wrong about that.

By 1941 the Soviet GRU had constructed a variety of agent nets targeting the German sea and railroad transport and communications arrangements through Sweden, and the key FRA courier, Allan Nyblad, worked for the GRU. Source "K" is listed as "Unidentified" in Venona T1463. Perhaps that really was the case at the GCHQ or NSA when this Venona message was solved, though that seems hard to believe, or perhaps it was just a gesture of some sort. Venona itself leaves little doubt about what kind of source "K" was. This fabulous GRU Stockholm source provided reports on a wide range of topics from transport shipping, to what is happening in the battlefields of Crimea, to the structure of small Finnish army units and what the Finns are building in the town of Kemi, to the German casualty reports by category by unit on the Northern front, and confirmation to the Russians that the Swedes were decrypting the messages of the Northern Russian fleet, etc., was of course nothing less than a major sigint source. This can be reasonably suspected already via an examination of other Venona telegrams from source "K", though a key source in Swedish Military HQ might have had the same kind of access in a secondary sense:

Number	Date	Area/Region Covered	Message Content Summary
883/41	25 Aug. 41	Shipping (Baltic?)	Re: German transport ships Würzburg and Urundi
884/41	25 Aug. 41	Naval?	Re: Karlskrona, is either the Swedish Navy base or the destroyer J8, also named Karlskrona.
908/41	28 Aug. 41	Denmark-Sweden straits	Communications Helsingör-Helsingborg and Copenhagen-Malmö.
914/41	29 Aug. 41	Unknown	Unclear
1404/41	21 Nov. 41	Kemi, Finnish town in Gulf of Bothnia	Some type of production is being organized by the Finns there.
1456/41	28 Nov. 41	Northern front (Polar, Salla, etc.)	Casualty, etc report on German and Finnish Army units in the region from front HQ to OKW (German General Staff) in Berlin
1546/41	10 Dec. 41	Kerch, Crimea	Mentions 1 Tank Army
1564/41	12 Dec. 41	Stockholm and Finland	Sigint./Army deployments. Mentions DF activity against USSR northern fleet, Swedes decrypting Russian cipher traffic, and provides a list of German divisions in Finland.
203/42	30 Jan. 42	Germany	A message concerning OKH the German Army High Command. Last message containing "K" (and a link to KARL)
520/42?¹	18 Mar. 42	Sweden	Covername KARL in HOSPITAL (=Prison)

Two relatively recent Swedish sources further assist in pinpointing Source "K". A particularly thorough review of the case was done by Wilhem Agrell, Professor in Intelligence Studies at Lund University (X). He unambiguously concluded that "K" matched the information provided in the interrogation records of the brothers Knut (Code Name EDGAR) and Allan Nyblad (9), one of whom was a courier at FRA and then obtained a duplicate key that could open the secure FRA briefcases. Bengt Beckman, a former FRA official also mentions the Nyblad case. (Y)

The brothers worked together and took out and photographed the contents in quick 10-15 minute working sessions on each courier delivery run. Then they handed over the film to Ivanov who gave it to Colonel N.I. Nikitushev of the GRU, cover name AKASTO, usually on the same day. And as quickly as possible the GRU encrypted the content in code sheets superencrypted with one-time pads and sent

them off to Moscow via radiotelegraphy. Very likely GRU Headquarters in Khodinka outside Moscow had the accounting of German losses not much later than OKW did in Berlin, or Swedish Army HQ in Stockholm. The GRU did not even bother breaking "K" up into several cryptonyms to hide the rather significant hints at a major sigint source, that's how secure they considered the one-time pad system. In fact, "K" was short for KARL, and it is probably not a coincidence that the courier worked out of an FRA building in Stockholm known as "KARLBO". So they really must have thought this was all completely secure. They were wrong about that.

Late in 1959 the NSA and GCHQ were scrounging around to obtain encrypted Russian messages from the 1940s that could be tested to see if they contained any of the reused one-time pad indicator numbers that would allow more messages to be "put in depth", and more text thus available to create solutions. In connection with this it was learned that the Swedes had intercepted quite a large number of Russian messages during World War II and kept them. They had also kept the messages sent via the radiotelegraphy of the Swedish PTT. So a deal was cut in which the Swedish messages were turned over to the GCHQ in return for a promise to get any solved content. Over the next 15 years the GCHQ and the NSA labored away on these messages and during some warm summer days in 1962 they managed to produce some breakthroughs in messages from 1941, including the 28th of November, that they possibly believed the Swedes would find extremely interesting. Evidence of a major sigint leak during the war.

By the time T1463 made its way back to Sweden in the final solved form sometime in the summer of 1962, it was about twenty years since the men behind the cryptonym "K" had been arrested and eventually convicted of the espionage at FRA. Aside from their confessions, and the astonishing range and depth of messages from "K", the very existence of the sources in the Venona "K" material match exactly their brief period of activity in 1941, and stops abruptly when the brother who was the courier was arrested in January, 1942. Later confirmed by message 520/42 to be "in hospital", GRU code for prison. Perhaps you think the arrest was the result of a long, painstaking and occasionally brilliant counterintelligence operation? It was not. He was arrested for performing illegal abortions, one of which went terribly wrong. When the GRU via the other brother tried to recruit a new courier, the attempt was reported and the whole thing unraveled during the criminal investigation that followed.

But the message was back again in Stockholm by mid-1962. Twenty plus years after they first decrypted the whole message, in the original German, likely even before the Germans in Berlin, the FRA got it back, fragments of it solved again. This time by the British working from Russian text. By then it had also no doubt been to the United States. And that's where it sits now, for all to see. Well traveled. Many hours of work dedicated to it. Twice encrypted, twice decrypted. Twice broken by cryptanalysis, and once stolen by spies. Translated at least twice and perhaps three times. Not bad for a routine casualty message from AOK Norwegen.

Venona, as I am sure you know, was an extremely secretive project at first known only to select individuals in US and UK sigint and counterintelligence organizations. President Truman was not informed and neither was the CIA until 1952. The only other organization that was well informed about the project already by 1945 was the KGB.

Background Material/Handout along with presentation at

T1463 below

USSR Ref. No: 3/NBF/T1463
Issued: 27/6/1962
Copy No: 201
"K's" INFORMATION: REPORT ON CERTAIN GERMAN AND FINNISH FORCES (1941)
From: STOCKHOLM
To: MOSCOW
No: 1456, 1457 28 Nov. 41
[Two-Part Message Complete]
[Part 1] To NORTHERN[SEVERNYJ].[i]
According to information from "K":[ii]

1. [2 groups unrecovered] northern front [1 group unrecovered] AOK NORWAY[ii]
from [1 group unrecovered] to [1 group unrecovered]:
XXXVI Army Corps[AK] [2 groups unrecovered] no information.
169 Infantry Division - 13, 3, 48, 185.
SS "NORD"[iv] - 53, [2 grps. 152, 26.
unrec.]
3 Mountain Division - 0, 19, 52, [1 grp.
unrec.]
2 Mountain Division - 0, 0, 9, 560.
[2 groups - [1 grp. 2, 193, 47.
unrecovered] Division. unrec.]
163 Infantry Division - [1 grp. 0, 27, 410.
unrec.]
6 Finnish Infantry Division - 2, 0, 27, 165,[a
Finnish Army Corps[AK] ([2 grps. - 135, 0, 614, 961.
unrecovered]) -
Distribution [Continued overleaf 2 3/NBF/T1463
(The first figure [1 group unrecovered] killed, the second missing, the
third wounded, the fourth sick).
[63 groups unrecovered]
([1 group unrecovered]) 124 north of KEMI. 4
[62 groups unrecovered]
[Part II]
[73 groups unrecovered]
...STER
[73 groups unrecovered]
[57 groups unrecoverable]
No. 471 AKASTO[v]
Notes: [a] Or : 265
Comments: [i] NORTHERN : MOSCOW addressee, so far only occurring in
messages sent in the STOCKHOLM lane.
[ii] K : Unidentified.
[iii] AOK NORWAY: German Army Headquarters in NORWAY
(Armeeoberkommando).
[iv] SS "NORD" : The name of an SS division in the German Army.
[v] AKASTO : Probably Col. N.I. NIKITUShev, Soviet Military
Attache in STOCKHOLM 1939-1945.